

ARNAB LAHA

Experimental Particle Physicist | Data Analysis, ML

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SUMMARY

Experimental particle physicist with over 6 years of experience in designing data analysis pipelines, processing TB scale data, cleaning, and drawing scientific conclusions using predictive modeling and statistical inference. Proficient in Python, C++, ML frameworks, and various distributed computing tools.

 **Analytical Thinking:** Develop algorithms to work with large datasets, improve identification algorithms to enhance the signal by 10%, and search for new particles.

 **Problem-Solving Skills:** Leverage data analysis techniques infused with physics information to address a few unsolved questions in physics using complex datasets.

 **Collaboration and Teamwork:** Worked in vibrant CMS and ATLAS collaboration of 3000 scientists. Actively contributed to projects with more than 100 collaborators to showcase the current status of the field, and mentored more than 10 master's, bachelor's, or summer students on their projects.

WORK EXPERIENCE

Post Doctoral Researcher

Nov 2025 - present

University of Witwatersrand, Johannesburg, South Africa

ATLAS Collaboration, CERN

Developing a data analysis strategy using the latest ATLAS dataset.

- Search for dark matter using novel and unconventional detector signatures.
- Simple cut and count-based algorithmic strategy to make the scientific output reinterpretable.
- Contributing to the analysis software development in an analysis team of more than 20 people.

Principal Analyst and PhD Student

Aug 2018 - Jul 2025

IISER Pune, Pune, India | CMS Collaboration, CERN

Thesis: **Search for vector-like leptons at CMS**

Developed efficient data analysis pipeline and techniques for large complex datasets.

- Processing and cleaning ≈ 100 TB scale of data to ease the analysis workflow, leveraging distributed computing like **HTCondor** and WLCG.
- Used **supervised ML** techniques to reduce noise by a factor of 1000 and derived scientific conclusions by **statistical analysis**.
- Implementation of **Generative AI** frameworks to solve physics problems, preparing a high-fidelity simulation chain.
- **VAE** or **GAN** (with dimension reduction twist) based generative models to produce a million instances of a physics process in a second.
- Principal code maintainer of the **ML in Physics** course offered at IISER Pune as a teaching assistant (taught unsupervised ML)–[Github](#), [Website](#).

Physics Object Expert

Sep 2021 - Jul 2025

CMS Collaboration, CERN

Reviewed more than 60 scientific papers as an internal physics object expert.

- Worked in a **management team** of 15 people, overseeing physics analyses and recommended multiple analysis teams across the globe to improve the usage of experimental physics objects, imple-

menting the **uncertainty** of identification algorithms.

- Regular exchange of recent developments, ideas, and **data issues** between the analysis overview team and object development team for efficient information percolation to end users.

EDUCATION

PhD in Physics	IISER Pune, Pune, India	2018 – 2025
Master's in Physics	University of Hyderabad, Hyderabad, India	2016 – 2018
Bachelor's in Physics	Presidency University, Kolkata, India	2013 – 2016

SELECTED PUBLICATIONS

- *CMS Thesis Endorsement*, Search for second and third generation vector-like leptons using muon and jets at $\sqrt{s} = 13$ TeV at the CMS experiment ,[Manuscript in preparation, combining more final states to achieve exclusion sensitivity](#).
- *The CMS Collaboration*, Review of searches for vector-like quarks, vector-like leptons, and heavy neutral leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV at the CMS experiment , [Physics Reports, Vol 1115](#)
- *The CMS Collaboration*, Inclusive nonresonant multilepton probes of new phenomena at $\sqrt{s} = 13$ TeV , [PhysRevD.105.112007](#)
- *The CMS Collaboration*, Search for dilepton resonances from decays of (pseudo)scalar bosons produced in association with a massive vector boson or top quark anti-top quark pair at $\sqrt{s} = 13$ TeV, [CMS-PAS-EXO-21-018](#)
- *Full publication list* in [INSPIRE-HEP](#)

SKILLS

Data analysis	ROOT, Pandas, NumPy
Language	Python, C++
ML Library	Scikit-learn, Tensorflow, Keras
Machine learning	Predictive Modeling, Supervised and Unsupervised ML, Clustering, Generative Models (VAE and GAN), Deep Neural Networks, Dimension reduction
Versioning	Github, Gitlab
Computing	HTCondor
Typesetting	Latex

AWARDS AND ACHIEVEMENTS

- 2025 [Breakthrough Prize](#) in Fundamental Physics as a member of the CMS Collaboration ([CMS Laureates](#)).
- CSIR NET (AIR 150) Junior Research Fellowship (2018-2020) and Senior Research Fellowship (2020-2023).
- *M.Sc. Gold Medal* in physics (CGPA: 9.16/10)
- AIR 23 (IIT-JAM 2017), State Rank 13 in class 10 exam (WBBSE 2011).

INTERESTS

✓Data Science, HEP x ML

Passionate about the potential of data science in solving complex problems (driven by physical laws of nature).

🧩Science Outreach

Enjoy teaching people about science, research, and the scientific temperament.